

Package ‘pediatric.zcalc’

September 30, 2026

Type Package

Title Z-Score Calculator for Biomarkers: Childhood to Young Adulthood

Version 0.1.1

Description Provides tools to compute individual percentile ranks and z-scores for clinical biomarkers in children, adolescents and young adults, based on age-, sex-, and height-specific reference data from the IDEFICS (Identification and prevention of Dietary and lifestyle-induced health EFfects In Children and infantS) study and the Biomarkers4Pediatrics collaboration. Supports the computation of a composite Metabolic Syndrome (MetS) score and associated monitoring/action levels for health monitoring. For more details see Ahrens et al. (2014) <[doi:10.1038/ijo.2014.130](https://doi.org/10.1038/ijo.2014.130)>.

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Encoding UTF-8

URL <https://github.com/bips-hb/pediatric.zcalc>

BugReports <https://github.com/bips-hb/pediatric.zcalc/issues>

LazyData true

Depends R (>= 3.5.0)

Imports gamlss.dist, pracma

Config/roxygen2/version 8.1.0

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

NeedsCompilation no

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Repository CRAN

Date/Publication 2026-09-30 09:40:02 UTC

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action_levels	<i>Compute Action Levels</i>
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Description

Assigns monitoring or action levels based on individual percentile ranks using standard IDEFICS thresholds.

Usage

```
action_levels(  
  df,  
  sex = NULL,  
  lvl_name = c("none", "monit", "action"),  
  perc_level = c(0.9, 0.95),  
  append = FALSE,  
  filter = NULL  
)
```

Arguments

df	A data frame containing percentile columns such as ‘waist_percentile’, ‘sbp_percentile’, ‘hdl_percentile’, etc.
sex	Character vector. Same length as ‘age’, ‘height’, and ‘values’. Accepts "f" for female or "m" for male.
lvl_name	Character vector of level labels. Defaults to ‘c("none", "monit", "action")’.
perc_level	Numeric vector of two percentiles used as cutoffs. Defaults to ‘c(0.9, 0.95)’ for 90th and 95th percentile.
append	Logical. If ‘TRUE’, appends action level columns to ‘df’. If ‘FALSE’, returns only the computed levels.
filter	Character. Optional. Limits calculation to a specific domain: “adiposity”, “blood_pressure”, “blood_lipids”, “blood_glu_insu”, or “overall”.

Details

Action levels are derived using ‘cut()’ on percentile values. For example, a value > 95th percentile maps to “action”. HDL is reversed (‘1 - hdl_percentile’) since low HDL values are considered unhealthy.

Value

A list (or a modified data frame if ‘append = TRUE’) containing action level classifications for each domain.

Examples

```
df <- data.frame(waist_percentile = c(0.85, 0.96))
action_levels(df)
```

```
df <- data.frame(
  hdl_percentile = c(0.1, 0.5),
  homa_percentile = c(0.4, 0.9),
  trig_percentile = c(0.6, 0.5),
  waist_percentile = c(0.9, 0.99),
  sbp_percentile = c(0.8, 0.01)
)
action_levels(df)
```

```
df <- data.frame(
  #sex = c("m", "m"),
  hdl_percentile = c(0.1, 0.5),
  homa_percentile = c(0.4, 0.9),
  trig_percentile = c(0.6, 0.5),
  crp_percentile = c(0.95, 0.9),
  waist_percentile = c(0.9, 0.99),
  sbp_percentile = c(0.8, 0.01)
)
action_levels(df, sex = c("m", "m"))
```

get_scores

Calculate Scores for Children and Young Adults

Description

Computes age-, sex- and (for ‘sbp’ and ‘dbp’) height-specific percentile ranks or z-scores for clinical biomarkers using IDEFICS study reference data and Biomarkers4Pediatrics collaboration (for ‘crp’).

Usage

```
get_scores(
  variable = "waist",
  sex = c("f", "m"),
  age = 6:5,
  height = NULL,
  values = c(20, 21),
  return_values = c("percentile", "z.score")
)
```

Arguments

variable	Character. The variable to assess. Must be one of the supported variables ("waist", "bmi", "hdl", "sbp", "dbp", "trg", "homa", "glu", "height", "insu", or "crp").
sex	Character vector. Same length as 'age', 'height', and 'values'. Accepts "f" for female or "m" for male.
age	Numeric vector. Ages of the children/young adults in years. Must be between 1 and 22.5 for 'crp', and between 2 and 11 otherwise.
height	Numeric vector or NULL. Required for height-dependent models ('sbp' and 'dbp'). Defaults to NULL.
values	Numeric vector. Observed values of the variable to score. Values must be supplied in the units used by the corresponding reference model: 'waist' and 'height' in cm; 'bmi' in kg/m ² ; 'sbp' and 'dbp' in mmHg; 'trg' and 'hdl' in mg/dL; 'glu' in mg/dL; 'insu' in μ IU/mL; 'homa' as dimensionless HOMA-IR; and 'crp' in mg/L.
return_values	Character vector. Specifies which outputs to return. Options include "percentile", "z.score".

Details

For triglycerides ('trg'), the original IDEFICS scoring procedure applies sex-specific percentile and z-score values at 45 mg/dL, corresponding to the lower measurement limit used in the IDEFICS study. These values are retained here to reproduce the original IDEFICS score calculation.

Value

A named list containing the requested scores. Each element is a numeric vector of the same length as 'values'.

Examples

```
get_scores(
  variable = "crp",
  sex = c("f", "f"),
  age = c(3, 12),
  values = c(4, 0.5)
)

get_scores(
  variable="dbp",
  sex=c("f", "m"),
  age=c(5,5),
  height=c(120,110),
  values=c(70,60)
)

get_scores(
  variable = "trg",
  sex = c("f", "m"),
```

```

    age = c(5.8, 6.5),
    values = c(50, 48) # mg/dL
  )

```

MetSScore

Calculate Metabolic Syndrome (MetS) Score

Description

Computes a composite MetS score from individual z-scores of clinical biomarkers.

Usage

```
MetSScore(df)
```

Arguments

df A data frame containing the following columns: 'waist_z.score', 'homa_z.score', 'sbp_z.score', 'dbp_z.score', 'trg_z.score', and 'hdl_z.score'. These are expected to be numeric vectors representing z-scores.

Details

The MetS score is computed as:

$\text{'waist_z} + \text{homa_z} + 0.5 \times (\text{sbp_z} + \text{dbp_z} + \text{trg_z} - \text{hdl_z})$

Value

A numeric vector representing the calculated MetS score for each row in the input data.

Examples

```

df <- data.frame(
  waist_z.score = c(1.2, 0.5),
  homa_z.score = c(0.8, 0.6),
  sbp_z.score = c(0.7, 0.3),
  dbp_z.score = c(0.6, 0.4),
  trg_z.score = c(1.0, 0.9),
  hdl_z.score = c(-0.5, -0.2)
)
MetSScore(df)

```

ScoreCalc

*Compute IDEFICS and B4P Scores for Multiple Variables***Description**

Applies `'get_scores()'` to multiple clinical biomarkers in a data frame and optionally returns MetS and cutoff levels.

Usage

```
ScoreCalc(
  df,
  return_input = FALSE,
  return_values = c("percentile", "z.score", "MetS", "cutoff.levels")
)
```

Arguments

<code>df</code>	A data frame with columns: <code>'sex'</code>, <code>'age'</code>, <code>'height'</code>, and observed values for any of the supported variables: <code>'bmi'</code>, <code>'glu'</code>, <code>'hdl'</code>, <code>'height'</code>, <code>'homa'</code>, <code>'insu'</code>, <code>'trg'</code>, <code>'waist'</code>, <code>'sbp'</code>, <code>'dbp'</code>, <code>'crp'</code>. Variables must be supplied using the units of the corresponding reference model: • <code>'age'</code>: years • <code>'height'</code>: cm • <code>'waist'</code>: cm • <code>'bmi'</code>: kg/m² • <code>'glu'</code>: mg/dL • <code>'insu'</code>: μIU/mL • <code>'homa'</code>: HOMA-IR (dimensionless) • <code>'hdl'</code>: mg/dL • <code>'trg'</code>: mg/dL • <code>'sbp'</code>, <code>'dbp'</code>: mmHg • <code>'crp'</code>: mg/L <code>'height'</code> is required for the height-dependent blood-pressure models (<code>'sbp'</code> and <code>'dbp'</code>).
<code>return_input</code>	Logical. If <code>'TRUE'</code> , includes original input columns in the result. Defaults to <code>'FALSE'</code> .
<code>return_values</code>	Character vector. Specifies which scores to compute. Options include <code>"percentile"</code> , <code>"z.score"</code> , <code>"MetS"</code> , and <code>"cutoff.levels"</code> .

Details

The function applies `'get_scores()'` to each recognized variable in the input and combines the results. If `"MetS"` is requested, a Metabolic Syndrome score is computed and optionally transformed to percentile ranks/z-scores.

Value

A data frame with score columns named as '<variable>_percentile', '<variable>_z.score', etc. Includes additional columns like 'MetS' or cutoff levels if requested.

Examples

```
df <- data.frame(  
  sex = c("f", "m"),  
  age = c(8, 15),  
  height = c(120, 125),  
  waist = c(55, 60),  
  homa = c(1.2, 1.4),  
  sbp = c(100, 105),  
  dbp = c(65, 70),  
  trig = c(80, 90), # mg/dL  
  crp = c(6, 2), # mg/L  
  hdl = c(50, 45) # mg/dL  
)
```

```
ScoreCalc(df, return_values = c("percentile", "cutoff.levels"))  
# for the 15 year old male everything except CRP is NA (outside IDEFICS age range)
```

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